Economic authorization and statistical certification

PARK–REVIVE research design note

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Status. Proposed extension following the economic-selection discussion. This note supplies a target distinction and an experiment design; it reports no new backtest, simulation result, power guarantee or frozen numerical protocol. The six-method, 500-market study already running retains its original configuration, seed membership and analysis plan.

1 The economic object is a learned policy

For a common pre-trade allocation b, define

$$C(w,b) = c\|w-b\|_1 + \frac{\eta}{2}(w-b)^\top \Lambda (w-b), \qquad \Lambda \succeq 0,$$

and realized quadratic utility

$$u(w,b;A) = w^\top A - \frac{\gamma}{2}(w^\top A)^2 - C(w,b).$$

With common true moments and the same feasible constraints, the optimized value

$$V_t(S;b) = \sup_{\mathrm{supp}(w)\subseteq S} \mathbb{E}[u(w,b;A_{t+1}) \mid \mathcal{F}_t]$$

cannot decrease when S expands: the larger feasible set includes the smaller one. This observation assumes that adding a permitted feature does not itself impose a separate fixed participation charge. It does not imply monotonicity of the out-of-sample values of estimated policies:

$$\Delta_t(B,S) = \mathbb{E}\big[u(\widehat{w}_t(B), b_t^B; A_{t+1}) - u(\widehat{w}_t(S), b_t^S; A_{t+1}) \mid \mathcal{F}_t\big].$$

Estimation error, changes in allocation and the two cost states matter. Even estimated optimized training value is monotone under nested supports; its positive increment alone is therefore weak protection against overfit.

The current full-study `Greedy` method is not a pure training-only economic selector. It fits allocations using the training history, but eligibility requires a positive accumulated OOS shadow-score mean, and annual priority ranks eligible moves by that mean. A method that initiates exposure from training estimates and estimation uncertainty would be an additional comparator.

2 Separate permission to trade from the tested null

A training-based forecast can authorize provisional exposure subject to a declared risk limit. An uncertainty adjustment must specify its estimation model, horizon and calibration before its new evaluation. A working posterior, bootstrap percentile or confidence bound does not acquire an OOS guarantee merely by being used as an economic gate.

For a predictable pair of shadow policies, an e-process addresses the sequential null that its score has nonpositive conditional mean throughout the comparison. At individual level a, crossing $1/a$ controls rejection under that null, assuming the score moment contract. The threshold is 20

when $a = .05$ for one comparison; using it repeatedly across features and episodes does not by itself control a global false-authorization probability.

Under the price-taking simulation contract, predictable provisional investment need not invalidate a shadow e-process that retains exactly the same policies, cost histories, filtration and valid moment bounds. Changing the reference, candidate, weights or cost state changes the tested object. The validity of the construction must then be checked for that object; previous capital cannot simply be relabeled as evidence for a different comparison. Neither a shadow rejection nor an economic forecast guarantees a positive future deployment gain in a changing market.

Certification against cash does not certify an increase

Suppose the current allocation is qw, $0 < q < 1$, and compare increasing to w against retaining qw from that same state. Put

$$m_1 = \mathbb{E}[w^\top A \mid \mathcal{F}_t], \quad m_2 = \mathbb{E}[(w^\top A)^2 \mid \mathcal{F}_t], \quad L_w = \|w\|_1, \quad K_w = w^\top \Lambda w.$$

The conditional utility gain from the increase is

$$\Delta_{\text{increase}} = (1-q)m_1 - \frac{\gamma}{2}(1-q^2)m_2 - c(1-q)L_w - \frac{\eta}{2}(1-q)^2 K_w. \tag{1}$$

The gain from entering w from cash is

$$\Delta_{\text{entry}} = m_1 - \frac{\gamma}{2}m_2 - cL_w - \frac{\eta}{2}K_w.$$

Consequently

$$\Delta_{\text{increase}} = (1-q)\Delta_{\text{entry}} - \frac{\gamma}{2}q(1-q)m_2 + \frac{\eta}{2}q(1-q)K_w.$$

For example, with no costs, $\gamma = 1$, $m_1 = .6$, $m_2 = 1$ and $q = .5$, full exposure has expected utility .1, half exposure has .175, and increasing exposure loses .075. These are admissible moments: a gross return with mean .6 and variance .64 has second moment 1.

Thus a certificate for full exposure versus cash does not establish that increasing a provisional position is beneficial. A favorable economic forecast for the increase must use (1), or its general state-dependent analogue. If the increase is to carry its own statistical certification, its candidate/reference score needs an appropriate e-process and error allocation. Keeping the original cash comparison is useful as a diagnostic, with its original interpretation.

3 A comparison that isolates the decision to wait

The full-study Tail-minus-Greedy contrast compares complete adaptive funds. It includes different selected supports, allocation states and later moves, so it does not isolate the effect of waiting. Add a controlled mechanism experiment with the following common objects:

1. Use the same training data and training-only candidate-selection rule in both arms. Define the candidate allocation path w_t and all subsequent fitting dates identically, without using either arm's wealth or trading state to change that common path.

2. Define a training uncertainty rule and a provisional fraction $q \in [0, q_{\max}]$, with $q_{\max} < 1$, before examining new results. A rejected economic proposal has $q = 0$ and remains in the experiment.

3. Run one common shadow candidate-versus-cash e-process using the declared score and its own shadow costs. Define a common, predictable full-exposure date from its crossing and any common economic gate. A decision made after a return is first tradable on the following date.

4. The waiting arm holds cash before that date. The provisional arm holds qw_t before it. Both target w_t afterwards. Each pays its own actual costs, including the different transition costs at the increase. The original shadow certificate remains a candidate-versus-cash certificate in both arms; it is not labeled a certificate of $q \to 1$.

5. Retain the complete evaluation horizon, including noncrossings, rejected training proposals and losses. Use paired markets with common shocks, and report separately the provisional losses and later outcomes.

This experiment identifies an exposure-timing contrast for the specified common candidate path and evidence clock. It does not establish the performance of a general adaptive feature-selection algorithm. A separate extension could test the increase directly, but that changes the evidence clock and needs its own design and guarantees.

Proposition 1 (Realized utility accounting)**.** *Let θ_t^P, θ_t^W be the provisional and waiting exposure fractions, $X_t = w_t^\top A_t$ the common full-target gross return, and C_t^P, C_t^W their actual trading costs. Their realized utility difference over H dates is exactly*

$$D_H = \sum_{t=1}^{H} \left[(\theta_t^P - \theta_t^W) X_t - \frac{\gamma}{2} \big((\theta_t^P)^2 - (\theta_t^W)^2 \big) X_t^2 - (C_t^P - C_t^W) \right].$$

Proof. Substitute $\theta_t^P w_t$ and $\theta_t^W w_t$ into the utility definition and subtract, retaining each arm's own cost state. □

With q fixed before the common deployment date T and full exposure in both arms afterwards, the gross-return and quadratic-risk terms cancel for $t \geq T$. Under the repository's cost convention, whose state is the previous target allocation, costs also coincide after the deployment date. Costs before and at T remain arm-specific. If $T > H$, including $T = \infty$, the pre-deployment contrast extends through all H dates. Wealth compounding is a distinct outcome and is not this additive utility identity.

4 Outputs and implementation boundary

The primary mechanism outcome should be the paired net utility difference, with expected-benefit, realized gross-return, quadratic-risk and trading-cost components retained separately. Risk outcomes should include negative provisional utility, downside outcomes, drawdowns and insolvency under the same wealth convention; none is a type-I error guarantee. Report waiting durations and noncrossings without conditioning performance on certification.

Before implementation and new evaluation, fix the uncertainty estimator and calibration, candidate-selection rule, provisional cap, investment horizon, review schedule, complete seed set, error allocation, outcome definitions and Monte Carlo analysis. The design follows observed pilot results and the user's proposal; it is not a pre-pilot registration. Reuse of existing market paths would be a paired extension, not additional independent market evidence.

No new numerical superiority claim follows from this note. In particular, the question *what benefit is forgone by waiting, and what downside is avoided?* requires the new controlled comparison; the existing adaptive fund contrast supplies complementary total-policy performance.

Implementation references inspected

`research/park_revive_full/selection.py`:
`Selector.refresh`, `eligible`, `choose`;
`research/park_revive_full/engine.py`:
OOS score updates and annual decisions.
`src/park_revive/policies.py`: previous-target cost state.
`src/park_revive/costs.py`: common utility and cost definitions.